60 DAYS | QUALITY ENGINEERING LAB

# DAY 01: COMPREHENSIVE QUALITY PERFORMANCE ANALYSIS OF MULTIPLE DEFECTS PER UNIT USING SIX SIGMA QUALITY METRICS

Laboratory Experiment & Industrial Training Manual

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|---|---|---|---|
| **Domain:** | Six Sigma & Process Excellence | **Industry:** | Electronics Manufacturing |
| **Product:** | Printed Circuit Board Assembly (PCBA) | **Prepared For:** | Quality & Mechanical Engineering Professionals |

## 1. AIM & OBJECTIVES

**AIM:** To conduct a comprehensive Six Sigma analysis of an electronics manufacturing process by evaluating defect occurrence, process yield, and overall quality performance for Printed Circuit Board Assemblies (PCBA).

### Key Objectives:

- Understand the fundamental concepts, philosophy, and methodology of Six Sigma quality engineering.
- Calculate Defects Per Unit (DPU), Defects Per Opportunity (DPO), and Defects Per Million Opportunities (DPMO).
- Analyse the impact of multiple defects occurring within individual product units.
- Evaluate process yield using First Pass Yield (FPY) across four core assembly stages.
- Calculate and interpret Rolled Throughput Yield (RTY) for the multi-stage manufacturing sequence.
- Calculate and interpret the Process Sigma Level with and without the conventional 1.5 Sigma Shift.
- Identify defect patterns using Pareto Analysis to prioritize quality improvement initiatives.
- Develop Corrective and Preventive Action (CAPA) frameworks to systematically eliminate manufacturing root causes.

## 2. PROBLEM STATEMENT & INDUSTRY BACKGROUND

**Problem Statement:** Modern high-density Printed Circuit Board Assemblies (PCBAs) contain hundreds of solder joints and components. Evaluating quality purely on a binary pass/fail basis conceals critical process variability. This experiment evaluates a production batch of 1,000 PCBAs where multiple defects can occur on a single unit. The analysis quantifies stage-wise yields, evaluates short-term vs. long-term Sigma capability, and establishes continuous improvement strategies.

### Electronics Manufacturing Process Stages (PCBA):

1. **Solder Paste Printing:** A mechanical stencil deposits solder paste onto copper PCB pads. Inspection ensures proper deposit volume and alignment to prevent bridging or starvation.

2. **Component Placement (Pick & Place):** High-speed automated surface-mount pick-and-place machines position components onto solder pads. Inspection verifies positional accuracy, component orientation, and presence.
3. **Reflow Soldering:** The board travels through a multi-zone thermal reflow oven where solder paste melts to establish mechanical and electrical bonds. Thermal profile control prevents tombstoning or cold joints.
4. **Final Inspection & Functional Testing:** Automated Optical Inspection (AOI), Automated X-ray Inspection (AXI), and functional test rigs perform final quality verification under operational voltages.

## 3. THEORETICAL BACKGROUND

Six Sigma is a data-driven quality framework targeting process variation reduction to achieve no more than 3.4 Defects Per Million Opportunities (DPMO). Formulated at Motorola and adopted globally, it provides standardized mathematical tools for evaluating manufacturing capability.

**Key Terminology Definitions**

- **Defect:** Any individual non-conformance meeting specified engineering limits (e.g., 1 solder bridge).
- **Defective Unit:** A product containing one or more defects (e.g., 1 PCB containing 3 distinct defects).
- **Defect Opportunity (O):** Total number of measurable potential failure points per unit.
- **Process Variation:** Natural (random) or assignable (special cause) deviations from target dimensions or values.

**The 1.5 Sigma Shift Convention:** Empirical industrial research indicates that over long production cycles, process means typically shift by approximately 1.5 standard deviations (1.5 σ) due to raw material batch changes, tool wear, environmental shifts, and operator variations. Therefore, a process exhibiting a short-term Z score of 6.0 σ corresponds to a long-term benchmark performance of 4.5 σ (3.4 DPMO).

## 4. MASTER FORMULA SHEET — SIX SIGMA QUALITY METRICS

### 1. Total Defect Opportunities (TOP)

$$\text{TOP} = \text{Total Units Produced} \times \text{Opportunities Per Unit}$$

Measures the total theoretical defect opportunities across the batch.

### 2. Defects Per Unit (DPU)

$$\text{DPU} = \text{Total Defects Found} / \text{Total Units Inspected}$$

Average number of defects observed per unit.

### 3. Defects Per Opportunity (DPO)

$$\text{DPO} = \text{Total Defects Found} / (\text{Total Units Inspected} \times \text{Opportunities Per Unit})$$

Probability of a defect occurring at any single opportunity.

### 4. Defects Per Million Opportunities (DPMO)

$$\text{DPMO} = \text{DPO} \times 1{,}000{,}000$$

Standardized metric scaling defect occurrence to a 1 million opportunity baseline.

### 5. First Pass Yield (FPY)

$$\text{FPY} = (\text{Units Passing First Attempt Without Rework} / \text{Total Units Entering Stage}) \times 100\%$$

Percentage of units passing a specific process stage without needing repair.

### 6. Rolled Throughput Yield (RTY)

$$\text{RTY} = \text{FPY}_{\text{Stage 1}} \times \text{FPY}_{\text{Stage 2}} \times \text{FPY}_{\text{Stage 3}} \times \text{FPY}_{\text{Stage 4}}$$

Probability of a unit passing through all sequential manufacturing stages defect-free.

### 7. Process Sigma Level (Without 1.5 Shift - Short-Term Z)

$$Z_{\text{unshifted}} = \text{NORMSINV}(1 - \text{DPO})$$

Where NORMSINV is the Inverse Standard Normal Cumulative Distribution Function.

### 8. Process Sigma Level (With 1.5 Shift - Long-Term Convention)

$$\text{Sigma Level (Shifted)} = Z_{\text{unshifted}} + 1.5$$

Standard industry benchmark rating adding the conventional 1.5 σ long-term shift.

## 5. INDUSTRIAL CASE STUDY DATASET

*Note: All dataset values below are simulated training data representing an electronics PCBA manufacturing batch.*

### Batch Parameters:

- **Total Inspected PCB Units (N):** 1,000 units

- **Defect Opportunities Per PCB (O):** 10 defined criteria per unit
- **Total Defect Opportunities (TOP):** 10,000 opportunities
- **Total Defects Observed (D):** 195 defects
- **Total Defective PCBs:** 135 PCBs (containing 1 or more defects)
- **Defect-Free PCBs:** 865 PCBs

### Table 1: Defect Category Breakdown

| Defect Category | Defect Count | Percentage (%) | Cumulative % |
|---|---|---|---|
| Soldering Quality Defects | 42 | 21.54% | 21.54% |
| Short Circuits & Solder Bridging | 35 | 17.95% | 39.49% |
| Component Misalignment | 28 | 14.36% | 53.85% |
| Insufficient Solder | 22 | 11.28% | 65.13% |
| Tombstoning & Component Damage | 18 | 9.23% | 74.36% |
| Missing Components | 15 | 7.69% | 82.05% |
| Excess Solder | 12 | 6.15% | 88.21% |
| Open Connections | 10 | 5.13% | 93.33% |
| Polarity Errors | 8 | 4.10% | 97.44% |
| Connector Fitting Defects | 5 | 2.56% | 100.00% |
| **TOTAL** | **195** | **100.00%** | **-** |

### Table 2: Stage-Wise Manufacturing Performance (FPY & RTY)

| Stage Description | Units Entering Stage | Passed First Attempt | Reworked / Failed | First Pass Yield (FPY) |
|---|---|---|---|---|
| Stage 1: Solder Paste Printing | 1,000 | 960 | 40 | 96.00% |
| Stage 2: Component Placement | 960 | 902.4 | 57.6 | 94.00% |
| Stage 3: Reflow Soldering | 902.4 | 830.2 | 72.2 | 92.00% |
| Stage 4: Final Inspection & Testing | 830.2 | 788.7 | 41.5 | 95.00% |

## 6. STEP-BY-STEP CALCULATIONS

### Metric 1: Defects Per Unit (DPU)

Formula: DPU = Total Defects / Total Units

Substitution: DPU = 195 / 1,000

Result: DPU = **0.195 defects/unit**

Interpretation: The manufacturing line produces an average of 0.195 defects per PCB board.

### Metric 2: Defects Per Opportunity (DPO)

Formula: DPO = Total Defects / (Total Units × Opportunities)

Substitution: DPO = 195 / (1,000 × 10) = 195 / 10,000

Result: DPO = **0.0195**

Interpretation: Each defect opportunity carries a 1.95% probability of failure.

### Metric 3: Defects Per Million Opportunities (DPMO)

Formula: DPMO = DPO × 1,000,000

Substitution: DPMO = 0.0195 × 1,000,000

Result: DPMO = **19,500 DPMO**

Interpretation: The process generates 19,500 defects for every 1 million opportunities.

### Metric 4: Yield and Defective Percentages

Defective Unit Percentage = (135 / 1,000) × 100% = **13.50%**

Defect-Free Unit Percentage = (865 / 1,000) × 100% = **86.50%**

### Metric 5: Rolled Throughput Yield (RTY)

Formula: RTY = $FPY_1 \times FPY_2 \times FPY_3 \times FPY_4$

Substitution: RTY = 0.9600 × 0.9400 × 0.9200 × 0.9500

Result: RTY = 0.78864 = **78.86%**

Interpretation: Only 78.86% of units pass through all four stages without requiring rework.

### Metric 6: Process Sigma Level Calculation

Unshifted Z (Short-Term): Z = NORMSINV(1 - 0.0195) = NORMSINV(0.9805) = **2.89 σ**

Shifted Sigma Level (Long-Term Convention): Sigma = Z + 1.5 = 2.89 + 1.5 = **4.39 σ**

## 7. GRAPHICAL ANALYSIS

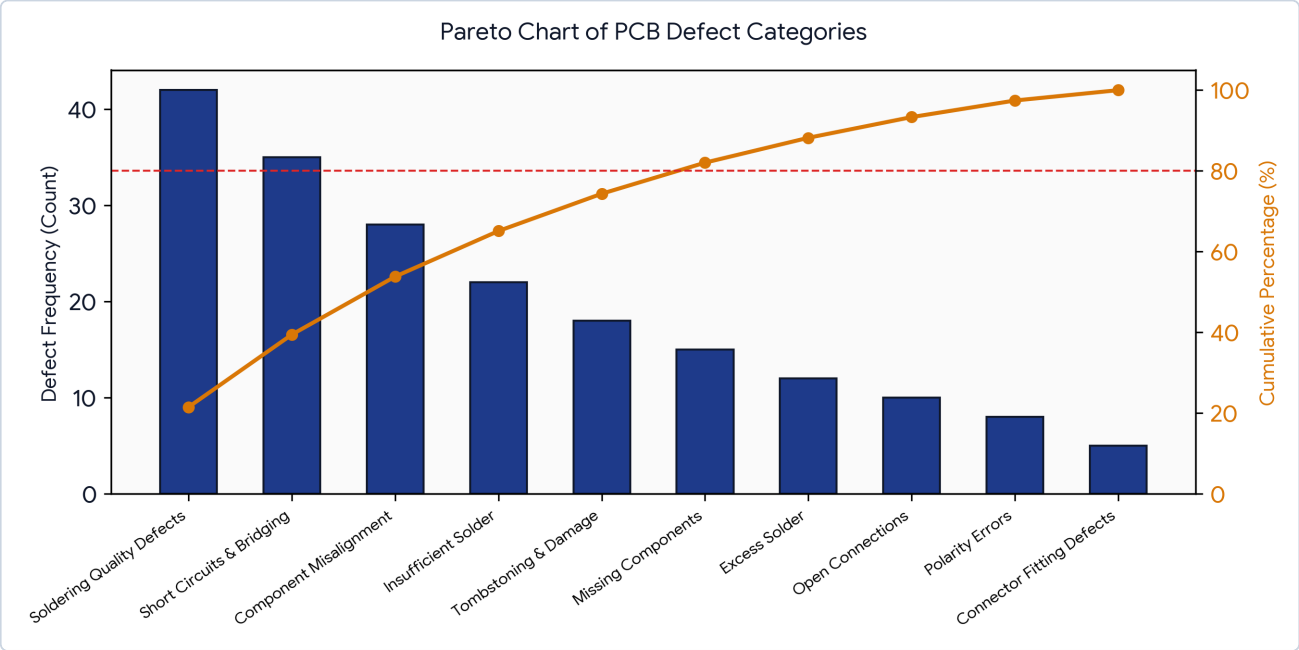



*Figure 1: Pareto Chart of PCB Defect Categories (Vital Few Defects Highlighted).*

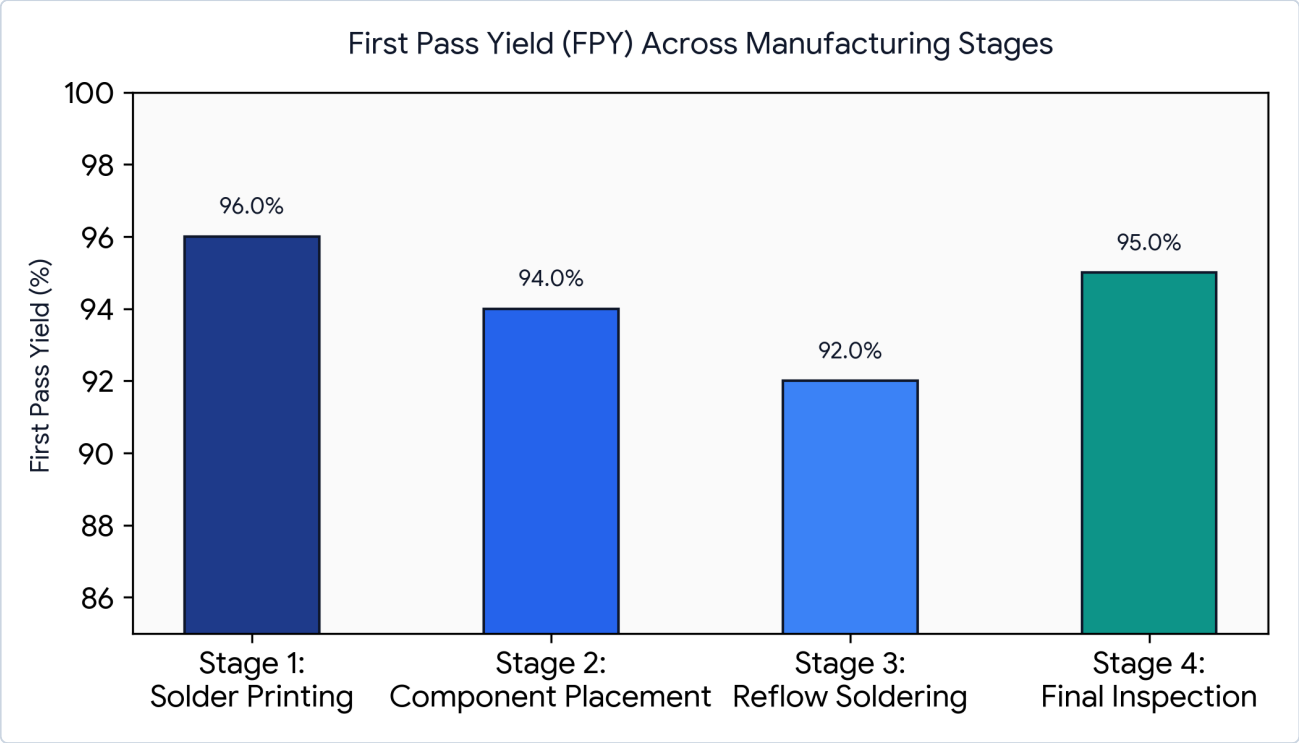



*Figure 2: First Pass Yield (FPY) Across Four Assembly Stages.*

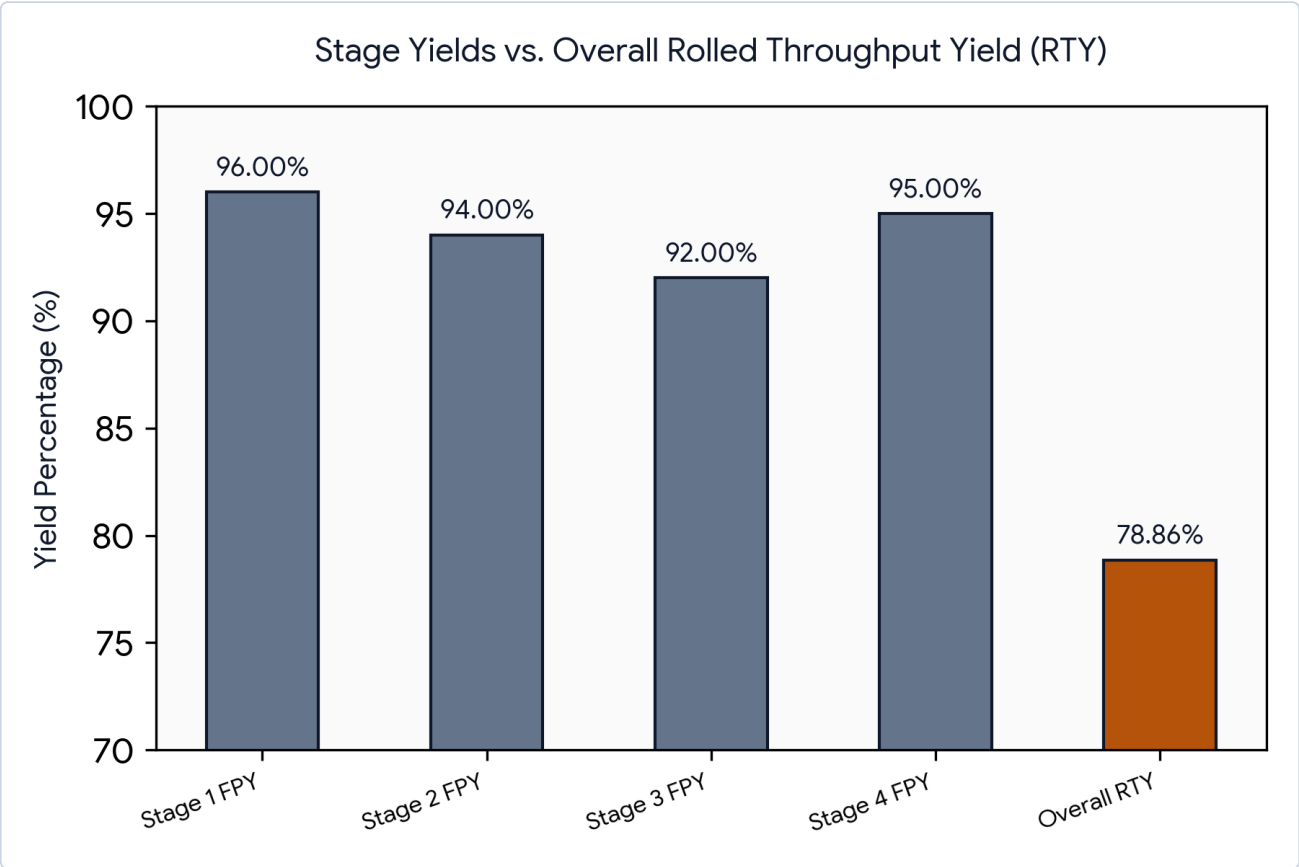



*Figure 3: Stage FPY Values vs. Overall Rolled Throughput Yield (RTY).*

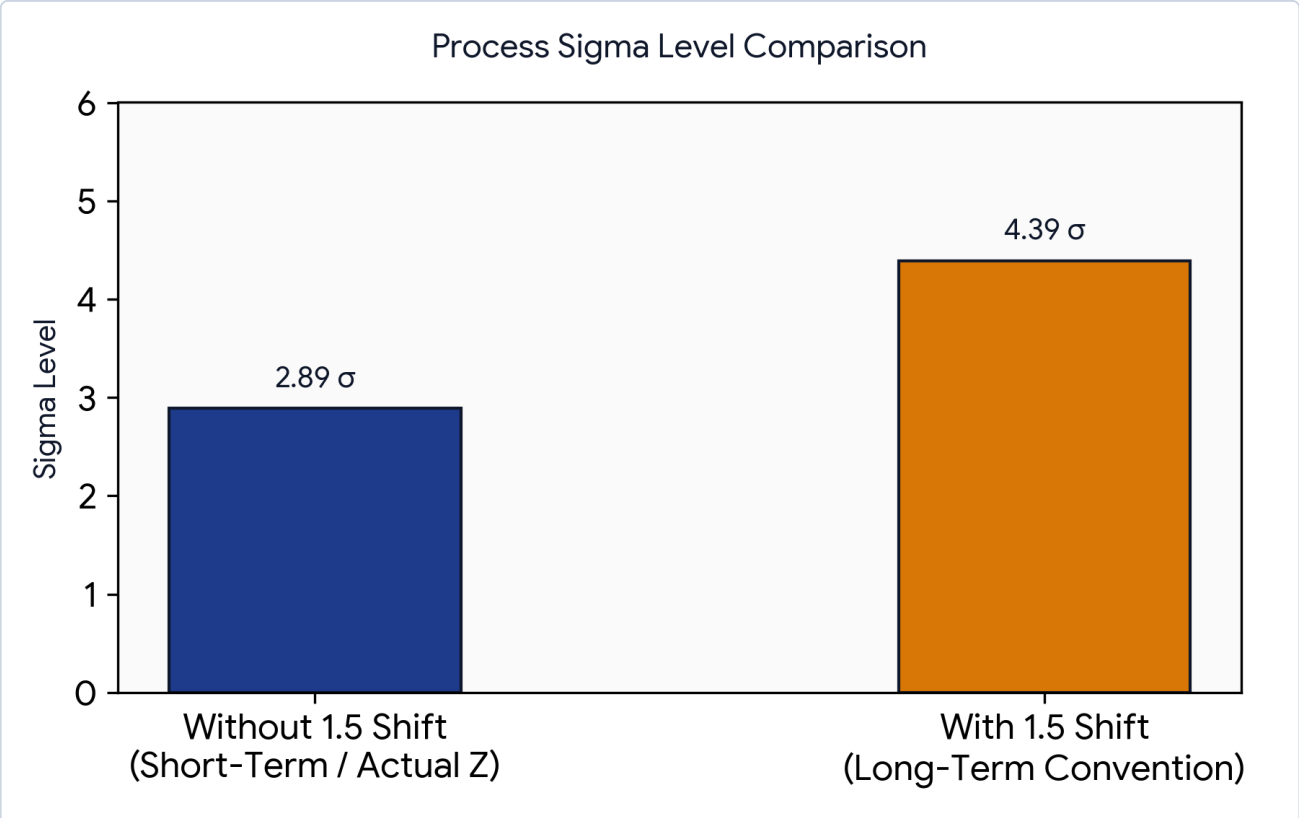



*Figure 4: Process Sigma Level (With vs. Without 1.5 Sigma Shift).*

## 8. COMPARATIVE ANALYSIS & INDUSTRIAL INTERPRETATION

| Quality Metric | Calculated Value | Engineering Evaluation & Action Trigger |
|---|---|---|
| Defects Per Unit (DPU) | 0.195 | Indicates high defect clustering across assembly operations. |
| DPMO | 19,500 | Requires significant continuous improvement to reach 6σ target (3.4 DPMO). |
| Stage 3 FPY (Lowest) | 92.00% | Reflow Soldering is the major process bottleneck. |
| Rolled Throughput Yield (RTY) | 78.86% | 21.14% of product incurs hidden factory rework costs. |
| Sigma Level (Short-Term Z) | 2.89 σ | True statistical capability based on short-term dataset. |
| Sigma Level (Shifted Benchmark) | 4.39 σ | Standard long-term industry benchmark rating. |

## 9. CORRECTIVE AND PREVENTIVE ACTION (CAPA) MATRIX

| Identified Problem | Possible Root Cause | Corrective Action (Immediate) | Preventive Action (Long-Term) | Verification Method |
|---|---|---|---|---|
| Soldering Quality & Short Circuits | Incorrect stencil thickness, worn squeegee blade, improper thermal profile. | Clean stencil immediately; adjust squeegee pressure and reflow profile. | Implement automated Solder Paste Inspection (SPI) before placement. | 3D SPI measurement logs & daily profile verification. |
| Component Misalignment | Nozzle wear, feeder vibration, vision system calibration drift. | Re-calibrate pick-and-place feeder and clean optical nozzles. | Schedule mandatory weekly nozzle maintenance & automatic vision calibration. | Pre-reflow AOI misalignment error rate tracking. |
| Tombstoning Defects | Uneven paste printing, unequal pad sizing, thermal imbalance. | Rework affected boards with micro-soldering station. | Redesign PCB footprint thermal relief pads according to IPC-7351. | Post-reflow AOI defect count log. |

## 10. TEST YOUR ENGINEERING SKILLS (PRACTICE NUMERICALS)

### Level 1 — Basic

**Q1: A batch of 500 PCBs has 15 total defects. Calculate DPU.**

Solution: DPU = 15 / 500 = **0.03 defects/unit**.

**Q2: A process produces 2,000 units with 5 opportunities per unit. Total defects found = 40. Calculate DPMO.**

Solution: DPO = 40 / (2,000 × 5) = 0.004. DPMO = 0.004 × 1,000,000 = **4,000 DPMO**.

### Level 2 — Intermediate

**Q3: A 3-stage process has FPY values of 98%, 95%, and 97%. Calculate RTY.**

Solution: RTY = 0.98 × 0.95 × 0.97 = 0.90307 = **90.31%**.

**Q4: If a PCB line has 20 defect opportunities per board and records 100 defects across 1,000 boards, calculate DPU and DPMO.**

Solution: DPU = 100 / 1,000 = **0.10**. DPO = 100 / (1,000 × 20) = 0.005. DPMO = **5,000 DPMO**.

### Level 3 — Advanced

**Q5: If a process has a calculated DPMO of 6,210, find the unshifted Z level and shifted Sigma Level.**

Solution: DPO = 0.00621. Z = NORMSINV(1 - 0.00621) = NORMSINV(0.99379) = **2.50 σ**. Shifted Sigma Level = 2.50 + 1.5 = **4.00 σ**.

## 11. VIVA VOCE QUESTIONS & ANSWERS

**Q1: What is the main difference between a defect and a defective unit?**

A: A defect is an individual non-conformance (e.g., 1 solder bridge), whereas a defective unit is an entire product containing 1 or more defects.

**Q2: Why is RTY a better metric than final yield after rework?**

A: RTY captures hidden factory rework costs across all stages, whereas final yield conceals scrap and repair labor.

**Q3: Why is 1.5 added to the short-term Z score in Six Sigma calculations?**

A: It accounts for long-term empirical process drift over extended production runs.

## 12. CONCLUSION & REFERENCES

**Conclusion:** In this experiment, evaluation of 1,000 PCBAs established a DPU of 0.195, a DPMO of 19,500, a Rolled Throughput Yield (RTY) of 78.86%, and a benchmark Process Sigma Level of 4.39 σ. Reflow Soldering was identified as the key bottleneck. Implementing the outlined CAPA matrix will eliminate top defect causes and advance line performance toward world-class Six Sigma standards.

### References:


- Montgomery, D. C. (2019). *Introduction to Statistical Quality Control* (8th ed.). John Wiley & Sons.
- Breyfogle, F. W. (2003). *Implementing Six Sigma: Smarter Solutions Using Statistical Methods* (2nd ed.). Wiley.
- IPC-A-610H (2020). *Acceptability of Electronic Assemblies*. IPC Association Connecting Electronics Industries.

## 13. FINAL REVISION SHEET (EXAM & INTERVIEW READY)

| Concept / Metric | Key Formula / Definition | Quick Benchmark / Rule of Thumb |
|---|---|---|
| DPU | Total Defects / Total Units | Average defect count per unit. |
| DPO | Total Defects / Total Opportunities | Probability of defect per opportunity. |
| DPMO | DPO × 1,000,000 | Six Sigma Standard: 3.4 DPMO = 6 σ. |
| RTY | Product of all Stage FPYs | Measures true rework-free process yield. |
| 1.5 σ Shift | Sigma Level = Z + 1.5 | Long-term empirical drift convention. |